$A_{TM} = \{\langle M, w\rangle \mid M \text{ is a TM that accepts } w\}$ Is A_{TM} recognizable?

U = "On input $\langle M, w\rangle$ where M is a TM and w is a string:

1. Run M on input w
2. If M accepts w, accept
   If M rejects w, reject "

If we run U on input $\langle M, w\rangle$

- If M accepts w, U will accept $\langle M, w\rangle$
- If M rejects w, U will reject $\langle M, w\rangle$
- If M loops indefinitely on w, U loops indefinitely on $\langle M, w\rangle$

A_{TM} is recognizable.

$A_{TM} = \{\langle M, w\rangle \mid M \text{ is a TM that accepts } w\}$ Show that A_{TM} is undecidable.

Assume that A_{TM} is decidable. There exists a TM H that decides A_{TM}

When we run H on input $\langle M, w\rangle$

- H accepts $\langle M, w\rangle \rightsquigarrow \langle M, w\rangle \in L(H) = A_{TM} \rightsquigarrow$ M accepts w

- H rejects $\langle M, w\rangle \rightsquigarrow \langle M, w\rangle \notin L(H) = A_{TM} \rightsquigarrow$ M does not accept w
  - M rejects w
  - M loops indefinitely on w

Consider this TM D.

D = "On input $\langle M \rangle$ where M is a TM:

1. Run H on input $\langle M, \langle M \rangle \rangle$ (string)
2. If H accepts $\langle M, \langle M \rangle \rangle$, reject
   If H rejects $\langle M, \langle M \rangle \rangle$, accept"

Suppose we ran TM D on input $\langle D \rangle$

- D accepts $\langle D \rangle$ $\rightsquigarrow$ H rejects $\langle D, \langle D \rangle \rangle$ $\rightsquigarrow$ $\langle D, \langle D \rangle \rangle \notin A_{TM}$
  D does not accept $\langle D \rangle$

- D rejects $\langle D \rangle$ $\rightsquigarrow$ H accepts $\langle D, \langle D \rangle \rangle$ $\rightsquigarrow$ $\langle D, \langle D \rangle \rangle \in A_{TM}$
  D accepts $\langle D \rangle$

A_{TM} is undecidable.

If A is recognizable and $\overline{A}$ is recognizable, A is decidable.

P $\wedge$ Q $\rightarrow$ R

Assume that A and $\overline{A}$ are recognizable. There exists TM M_A that recognizes A and $M_{\overline{A}}$ that recognizes $\overline{A}$.

M' = "On input w:

1. Run M_A and $M_{\overline{A}}$ on input w simultaneously
2. If M_A accepts w, accept
   If $M_{\overline{A}}$ accepts w, reject"

$(P \wedge Q) \rightarrow R \equiv \neg(P \wedge Q) \vee R \equiv (\neg P \vee \neg Q) \vee R \equiv \neg P \vee (\neg Q \vee R) \equiv \neg P \vee (R \vee \neg Q) \equiv$

$(\neg P \vee R) \vee \neg Q \equiv \neg(P \wedge \neg R) \vee \neg Q \equiv (P \wedge \neg R) \rightarrow \neg Q$

If A is recognizable and A is undecidable, $\overline{A}$ is unrecognizable.

A_{TM} A_{TM} $\overline{A_{TM}}$

$\Sigma = \{0, 1\}$

M_1 = "On input x:
1. accept"

$L(M_1) = \Sigma^*$
M_1 halts on all strings.

M_2 = "On input x:
1. reject"

$L(M_2) = \emptyset$
M_2 halts on all strings.

M_3 = "On input x:
1. Enter infinite loop"

$L(M_3) = \emptyset$
M_3 loops indefinitely on all strings.

M_4 = "On input x:
1. If x == 01101, accept
2. reject"

$L(M_4) = \{01101\}$

Consider a TM M and a string w

N_1 = "On input x:
1. Run M on input w
2. If M accepts w, accept
   If M rejects w, reject"

$L(N_1) = ?$ don't know

If M accepts w, $L(N_1) = \Sigma^*$

If M rejects w, $L(N_1) = \emptyset$

If M loops indefinitely on w, $L(N_1) = \emptyset$

↳ N_1 loops indefinitely on all strings.

$HALT_{TM} = \{\langle M, w\rangle \mid M \text{ is a TM that halts on input } w\}$ * Prove that $HALT_{TM}$ is undecidable.

Assume that $HALT_{TM}$ is decidable. There exists a TM R that decides $HALT_{TM}$.

Construct TM S that decides $A_{TM} = \{\langle M, w\rangle \mid M \text{ is a TM that accepts } w\}$

S = "On input $\langle M, w\rangle$ where M is a TM and w is a string:

1. Construct TM M' as follows:
   M' = "On input x:
   1. Run M on input w
   2. If M accepts w, accept
      If M rejects w, enter infinite loop"
2. Run R on input $\langle M', 011\rangle$
3. If R accepts $\langle M', 011\rangle$, accept
   If R rejects $\langle M', 011\rangle$, reject"

- M accepts w → M' accepts all strings
  M' halts on all strings
  $\langle M', x\rangle \in HALT_{TM}$ for any string x.
- M rejects w → M' loops indefinitely on all strings
  $\langle M', x\rangle \notin HALT_{TM}$ for any strings x.
- M loops indefinitely on w
  ↓
  M' loops indefinitely on all strings.
  $\langle M', x\rangle \notin HALT_{TM}$ for any string x.